Parallel circuits

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Relate branch currents to a shared potential difference.

SUCCESS CRITERIA - I can...

1. Identify branches connected across the same two nodes.
2. Use equal voltage across parallel branches.
3. Add branch currents to find supply current.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A current of 3 A flows for 5 s. Find the charge that passes.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $1/R_{total} = 1/R_1 + 1/R_2$; $I_{total} = I_1 + I_2$; $V_1 = V_2$. Parallel branches connect to the same two nodes and share voltage.

Worked reasoning: Two parallel branches of 6 and 3 ohms each have 6 V across them. Branch currents are 1 and 2 A; source current is 3 A.

Your turn: Draw the two branches and mark where the currents split and recombine.

Marzano check - circle: ① ② ③

03 Look and explain

Mark the two shared nodes in the parallel diagram. If one branch opens, can the other remain powered? Explain.



Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

04 Build the solution

Two branches each have 6 V. Their resistances are 3 Ω and 6 Ω. Find currents using I=V÷R, then add them.

Marzano check - circle: ① ② ③

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05 Use the idea

Why can the equivalent resistance be smaller than either branch resistance?

Marzano check - circle: 1 2 3

06 Choose a method

A bundle of parallel rays passes through a lens and becomes 4 times as wide farther beyond it, with rays spreading immediately after the lens. Identify the lens type and sketch the rays.

Marzano check - circle: 1 2 3

07 Exit check

Two parallel branches draw .2 A and .3 A. Find the current supplied to their junction.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Return to the worked example in problem 2. Add a third 6 ohm branch to this ideal supply. Predict each existing branch current and the new total current.

Marzano check - circle: 1 2 3